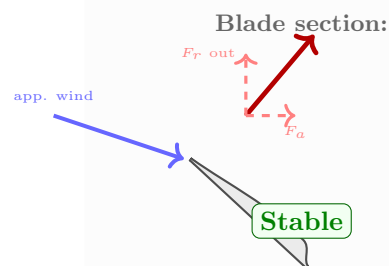
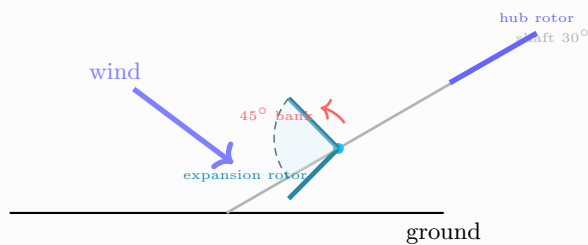


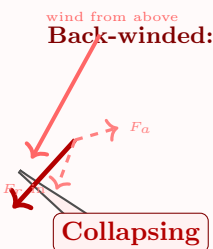
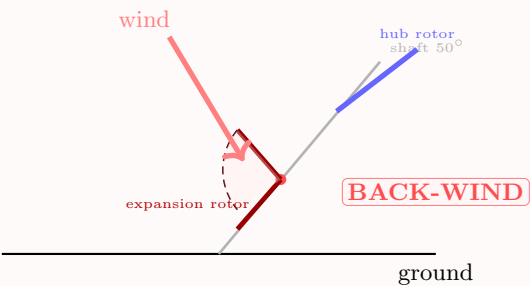
Expansion Rotor Bank Angle & Pitch Depower Risk

Normal: Rotor Sweeps Truncated Cone



- Expansion rotor:**
- Hub at ring position on shaft
 - Blades banked 45° outward
 - Sweep hollow truncated cone
 - Lift $\rightarrow F_r$ (spreads ring)
 - $+ F_a$ (shaft compression)

Pitch Depower: Back-Wind Risk



- Back-wind mechanism:**
- Shaft tilts up during depower
 - Blade orientation unchanged
 - Apparent wind hits from above
 - Lift reverses $\rightarrow F_r$ inward
 - r_{eff} drops, shaft buckles

Static DE optimiser cannot detect transient back-wind collapse — dynamic ODE validation required.

Mitigation: lower bank angle (20–30°), symmetric airfoils, conservative prototyping until validated